Syntax reference

Octave and Python (NumPy) — MATH 107

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Pick **one** language for the term. Lab hubs show both; this page is the growing cheat sheet. Full docs: Lachniet PDF and NumPy links on Resources.

Indexing reminder: Boyd's x_1 is the first entry. Octave is **1-based** (`x(1)`). Python / NumPy is **0-based** (`x[0]`).

Vectors and basics

Task	Octave	Python (NumPy)
Column vector	`x = [1; 2; 3]`	`x = np.array([1.0, 2.0, 3.0])`
Length n	`length(x)`	`len(x)` or `x.size`
First / second entry	`x(1)`, `x(2)`	`x[0]`, `x[1]`
Add / scale	`x+y`, `2*x`	`x+y`, `2*x`
Dot product $x^\mathsf{T} y$	`x'*y`	`x @ y` or `np.dot(x,y)`
Plot in 2D	`plot(x(1), x(2), 'o')`	`plt.scatter(x[0], x[1])` (after `import matplotlib.pyplot as plt`)

Norm, distance, angle

Task	Octave	Python (NumPy)
Norm $\|x\|$	`norm(x)`	`np.linalg.norm(x)`
Distance $\|x-y\|$	`norm(x-y)`	`np.linalg.norm(x-y)`
Cosine similarity	`(a'*b)/(norm(a)*norm(b))`	`(a @ b)/(np.linalg.norm(a)*np.linalg.norm(b))`
Angle (degrees)	`acosd(cos_th)`	`np.degrees(np.arccos(np.clip(cos_th,-1,1)))`

Many candidates (rows of a matrix)

Store each candidate as a **row** of `Z`. Query `x` is a column in Octave examples below (transpose with `x'` when subtracting from rows).

Task	Octave	Python (NumPy)
Stack candidates as rows	`Z = [1 2; 3 4; 5 6]`	`Z = np.array([[1,2],[3,4],[5,6]])`
All distances to `x`	`d = sqrt(sum((Z - x').^2, 2))`	`d = np.linalg.norm(Z - x, axis=1)`
Nearest index	`[~, j] = min(d)` → **1-based**	`j = np.argmin(d)` → **0-based**
Smallest distance	`min(d)`	`np.min(d)`

Matrices and $Ax = b$

`b` is a column. In Python, `*` is entrywise; matrix multiplication is `@`. Run `import numpy as np` first.

Task	Octave	Python (NumPy)
Enter a matrix	`A = [1 2; 3 4]`	`A = np.array([[1., 2],[3, 4]])`
Column b	`b = [5; 6]`	`b = np.array([5., 6])`
Product Ax	`A * x`	`A @ x`
Augmented $[A \mid b]$	`M = [A b]`	`M = np.column_stack((A, b))`
Left-multiply by E	`M = E * M`	`M = E @ M`
Unique solve	`x = A \ b`	`x = np.linalg.solve(A, b)`
Check $Ax - b$	`A * x - b`	`A @ x - b`

Unit I labs using this sheet

Lab	Focus
Lab 0	Enter vectors, arithmetic, basic plot
Lab 1	Distances and nearest neighbor
Lab 2	Gaussian elimination, augmented matrices, `A\b` / `solve`
Lab 3	Solve $Ax = b$, residual
Lab 4	Iterate $x \leftarrow Ax$
Lab 5	Eigenpairs / steady state
Lab 6	Least squares and truncated SVD